[NASA SUITS]

Software Design Review

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Meet the Team!

- Israel Cantu [2nd Year B.S CS]
- Cameron Huang [2nd Year B.S CS]
- Sumodha Pokhrel [2nd Year B.S CS]
- Abby West [2nd Year M.A. Human-Computer Interaction & Human Factors]
- Evelyn Bang [2nd Year M.A. Human-Computer Interaction & Human Factors]
- Will Ingersoll [3rd year B.A COGS]
- Steven Zhang [2nd year B.S CS]
- Frank Liu [2nd year B.S CS & Business]
- Isabella Alvarez [1st year B.S CS]
- Diego Sotomayor [1st year B.S CS]
- Aurial Teemoi [1st year B.S CS]
- Lilian Tsai [1st year B.S COGS]
- Isadora Knudsen [1st year B.S ECE]

CONCEPT OF OPERATIONS

2 screens!

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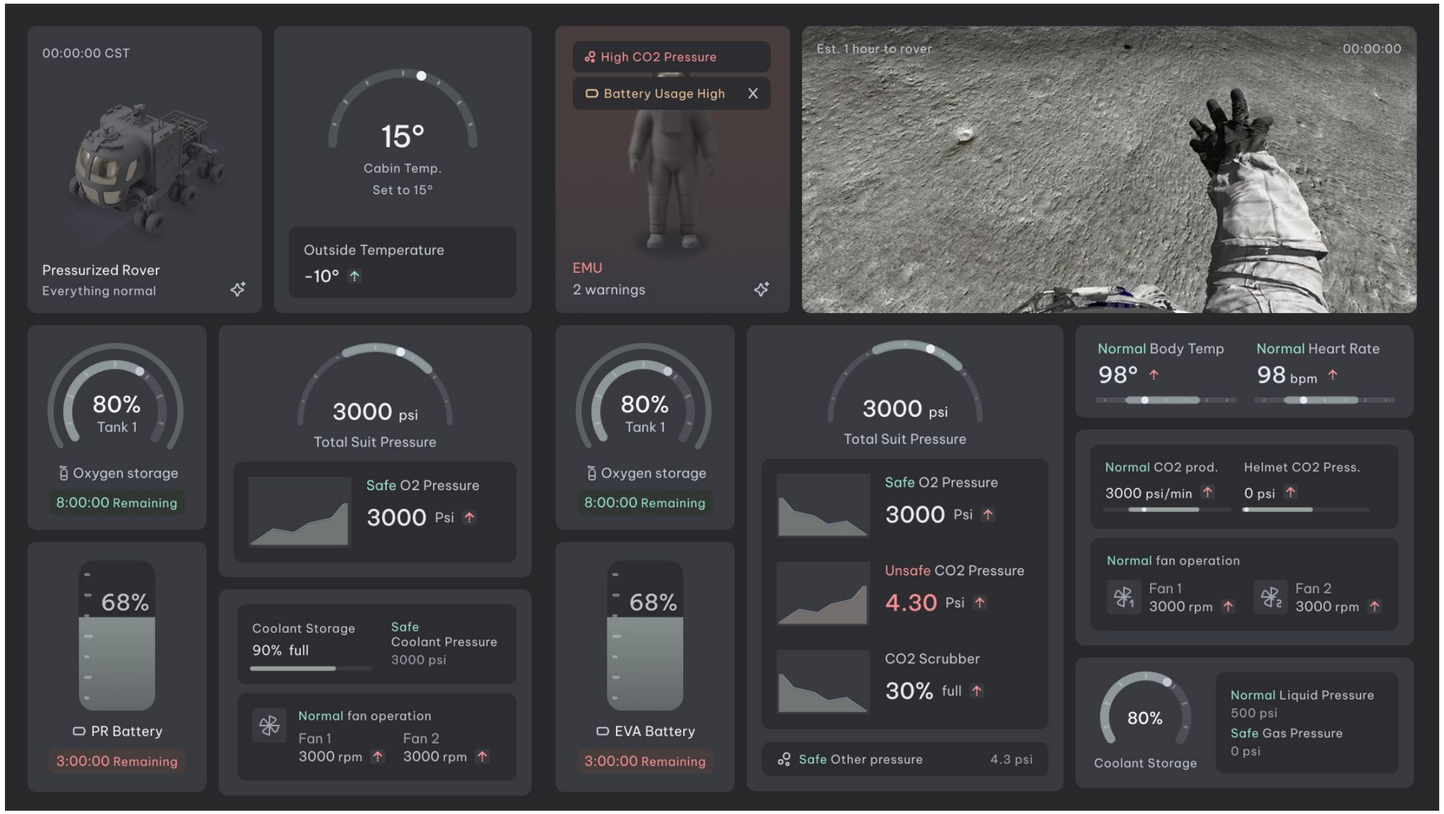
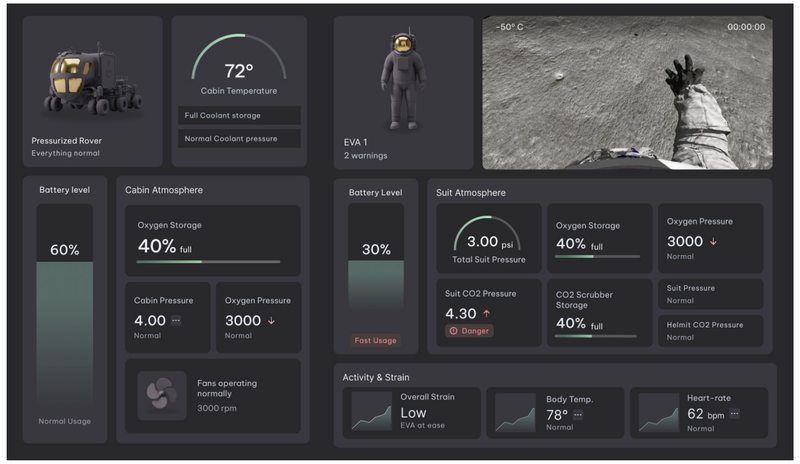
Data Dashboard Screen

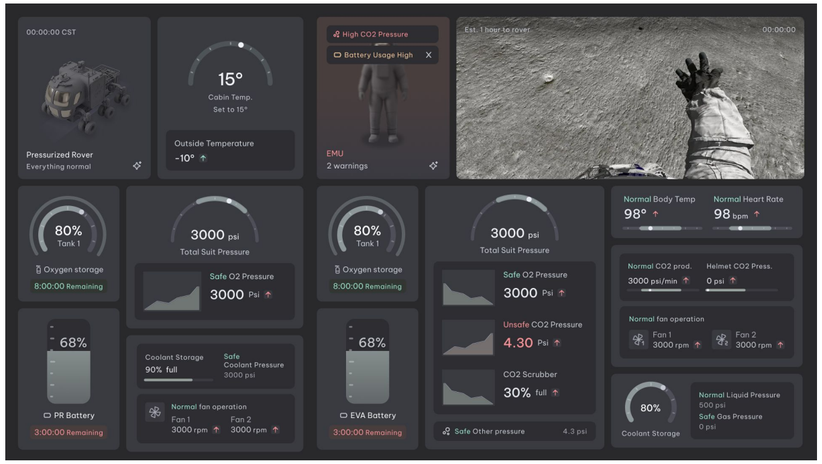


- Alerts and warnings
- Life support systems (O_2, CO_2, pressure, coolant)
- Astronaut health monitoring
- Power and resource levels
- Real-time system graphs and gauges
- AI assistant status updates

Data Dashboard - Iterations



4 iterations

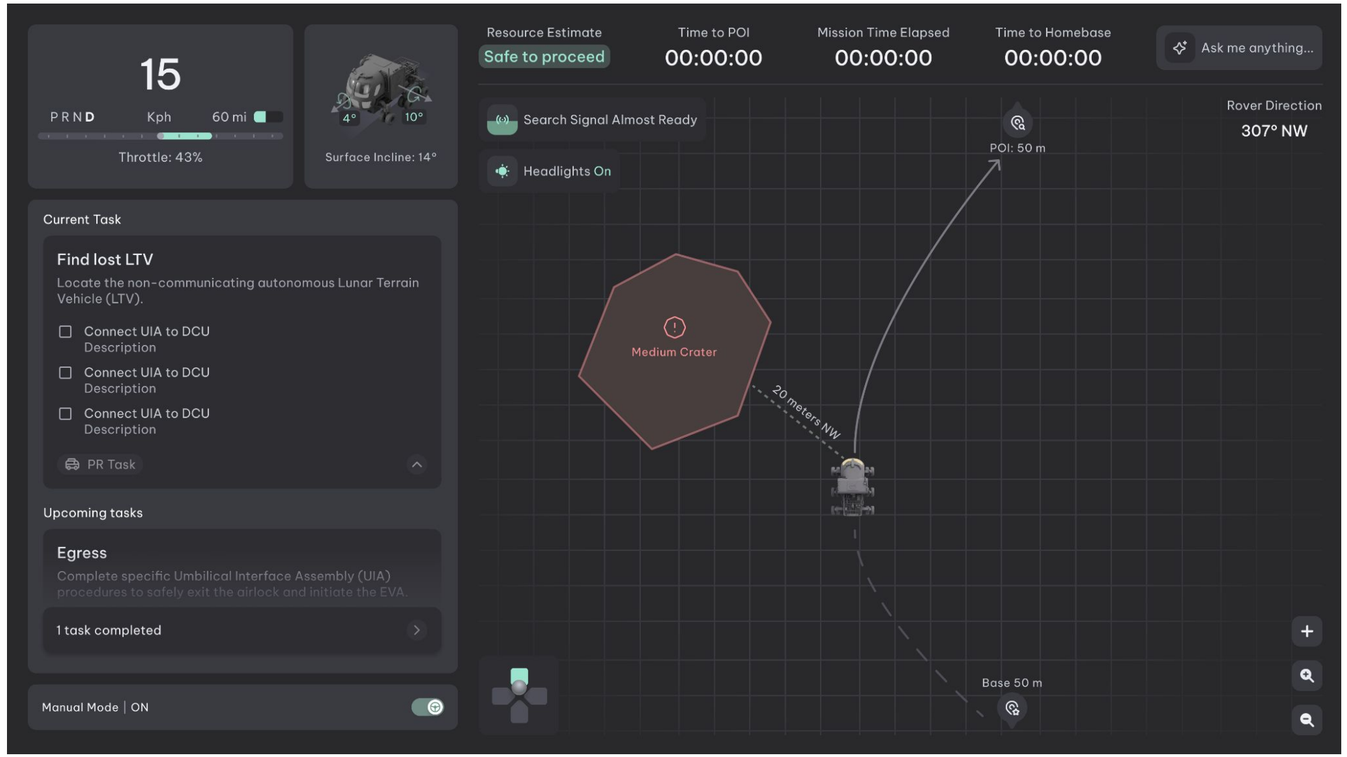


- Determined consistent placement for alerts at the top of the navigation screen
- Added more operational data while simplifying system displays (e.g., liquid and gas pressure)
- Added graphs to show trends for oxygen tanks and scrubbers
- Introduced tick indicators for quick confirmation of system status
- Reduced color intensity so critical alerts stand out more clearly

Data Dashboard - Requirements Fulfilled

Subsystem	SUITS Req IDs	What it satisifies
Spacesuit Telemetry	PR-2a (*May*) PR-2b (*Shall*)	Displays biomedical data (heart rate, body temp, CO_2 production) and EVA suit state (battery, pressure, O_2). Easily accessible at all times
Rover Telemetry	PR-2b (*Shall*)	Rover cabin state presented alongside EVA suit data in a unified dashboard view.
Caution & Warning	PR-5e-i (*Shall*) PR-5e-ii (*Shall*) PR-7c (*Shall*)	C&W system alerts crew when telemetry enters off-nominal ranges; AIA recommends corrective procedures in real time.
AIA Status Updates	PR-7d (*Shall*)	AIA delivers concise natural-language telemetry readouts via text and voice (e.g., “Primary O_2 47%, Secondary O_2 99%”).

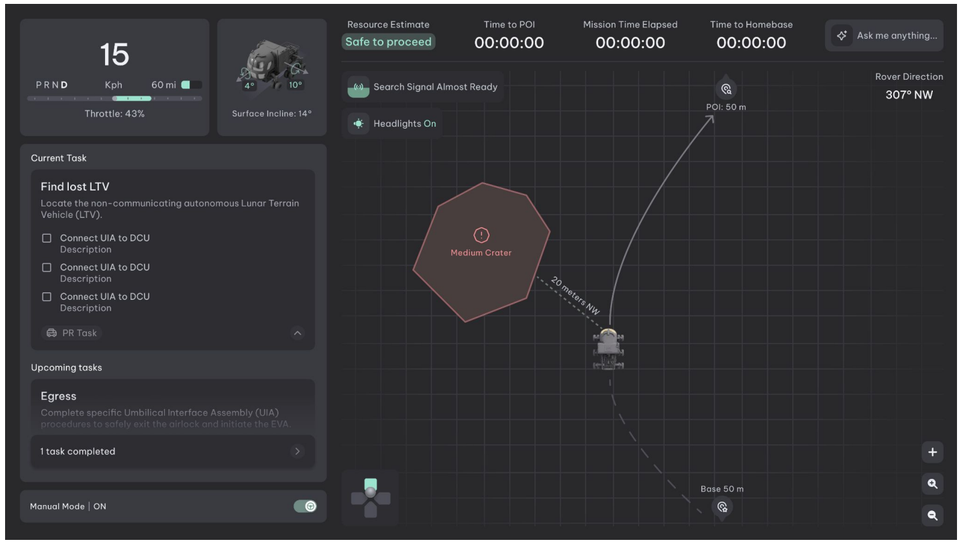
Navigation Screen



- Real-time map for navigation and asset tracking
- Current task and upcoming task display
- Key navigation metrics (speed, direction, distance, time)
- Terrain awareness indicators (pitch, roll, surface incline)
- Resource estimate to support safe mission decisions
- Alerts and Status Indicators

Navigation - Iterations

3 iterations



- Decluttered interface to improve readability and focus
- Created dedicated space for task visibility and workflow tracking
- Added 3D pitch and roll indicators for terrain awareness
- Introduced autonomous navigation control button
- Emphasized rover speed for real-time driving awareness

Navigation - Requirements Fulfilled

Subsystem	SUITS Req. IDs	What it satisifies
Live 2D Map	PR-5a (*Shall*) PR-5b-i (*May*) PR-5b-ii (*May*) PR-6d (*Shall*)	Real-time map tracks surface assets (rover, EV, LTV); draws EV traversed paths and PR traversed+projected paths.
LTV Search & Beacon	PR-3a (*Shall*) PR-5c-i (*Shall*) PR-5c-ii (*Shall*) PR-5c-iii (*Shall*)	LTV beacon data; search radius from last-known location + consumable limits; dynamic search pattern from LIDAR/beacon input
POIs & Waypoints	PR-5d-i (*Shall*) PR-5d-ii (*Shall*)	User can drop and label pins on map. Bidirectional pin sync between PR ↔ EV interface also possible

Navigation - Requirements Fulfilled (cont.)

Subsystem	SUITS Req. IDs	What it satisifies
Best-Path & Hazard Avoidance	PR-6a (*Shall*) PR-6b (*Shall*) PR-6c (*Shall*)	Autonomous routing determines optimal path; avoids boulders (≤1m) and craters (≤2m); replans autonomously
Rover Health & Asset Tracking	PR-6e (*Shall*) PR-7a (*Shall*) PR-7b (*Shall*)	Tracks PR speed, heading, and orientation; monitors life support health; provides predictive resource analytics and turnaround point estimates.
Tasks	PR-4a (*Shall*) PR-5f-i (*Shall*) PR-5f-ii (*Shall*)	Displays completed and upcoming EVA procedures; shows mission elapsed time (HH:MM:SS) per NASA timer standards
Rover Control	PR-1a (*Shall*)	DUST Rover control via Xbox controller or on-screen controls

SOFTWARE DEVELOPMENT PLAN

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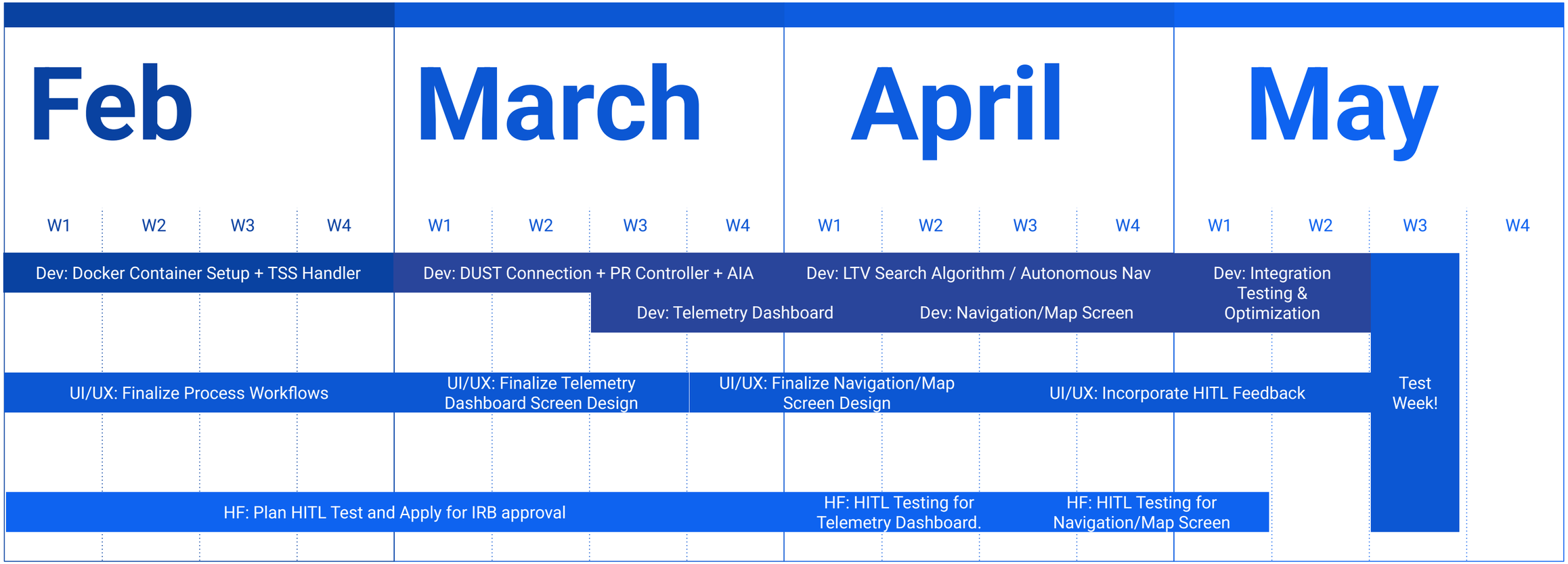
Development Timeline

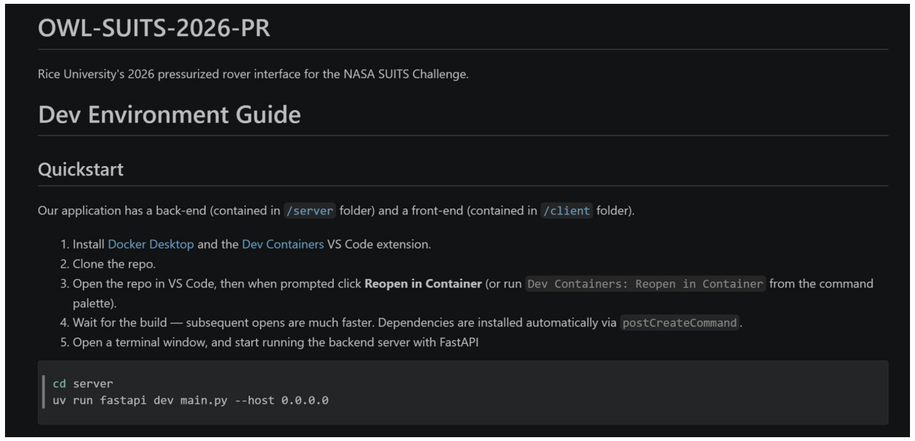


Code Management

- Version control managed via **GitHub**
- **GitHub Projects** for task tracking and code specific project management
- **Docker container** ensures a **unified development environmen**t across all team members
- Coding style standards enforced through **linting**
- **CI/CD** pipeline planned for **automated testing and deployment**

Documentation

- **Frontend:** React components documented with **Storybook**
- **Backend:** FastAPI automatically generates and hosts interactive **API documentation**
- **Codebase:** Markdown developer guides maintained in a dedicated **/docs** folder
- **User Guide:** Detailed end-user guide covering full application usage
- **Team:** Secrets, environment configs, and dev-specific information securely shared via **Coda**



OWL-SUITS-2026-PR

Rice University's 2026 pressurized rover interface for the NASA SUITS Challenge.

Dev Environment Guide

Quickstart

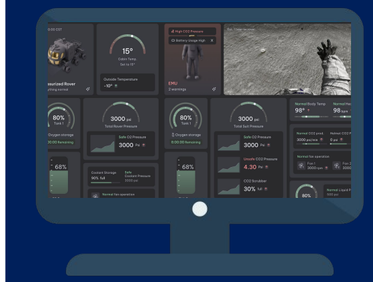
Our application has a back-end (contained in `/server` folder) and a front-end (contained in `/client` folder).

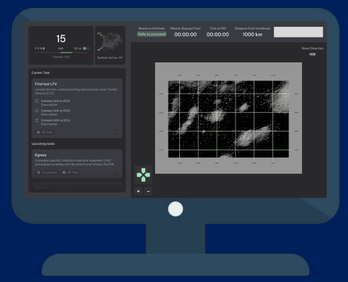
1. Install Docker Desktop and the Dev Containers VS Code extension.
2. Clone the repo.
3. Open the repo in VS Code, then when prompted click **Reopen in Container** (or run `Dev Containers: Reopen in Container` from the command palette).
4. Wait for the build — subsequent opens are much faster. Dependencies are installed automatically via `postCreateCommand`.
5. Open a terminal window, and start running the backend server with FastAPI

```
cd server
uv run fastapi dev main.py --host 0.0.0.0
```

Architecture & Design

User Interaction





Backend Systems

AIA

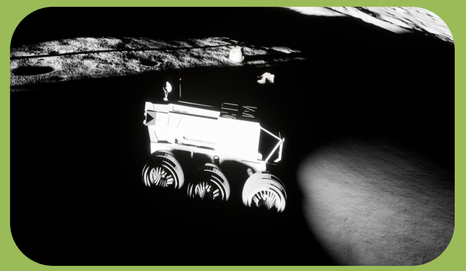
Server

mongo DB

Database

Outside Services

TSS



DUST

EMBEDDED DEVICES

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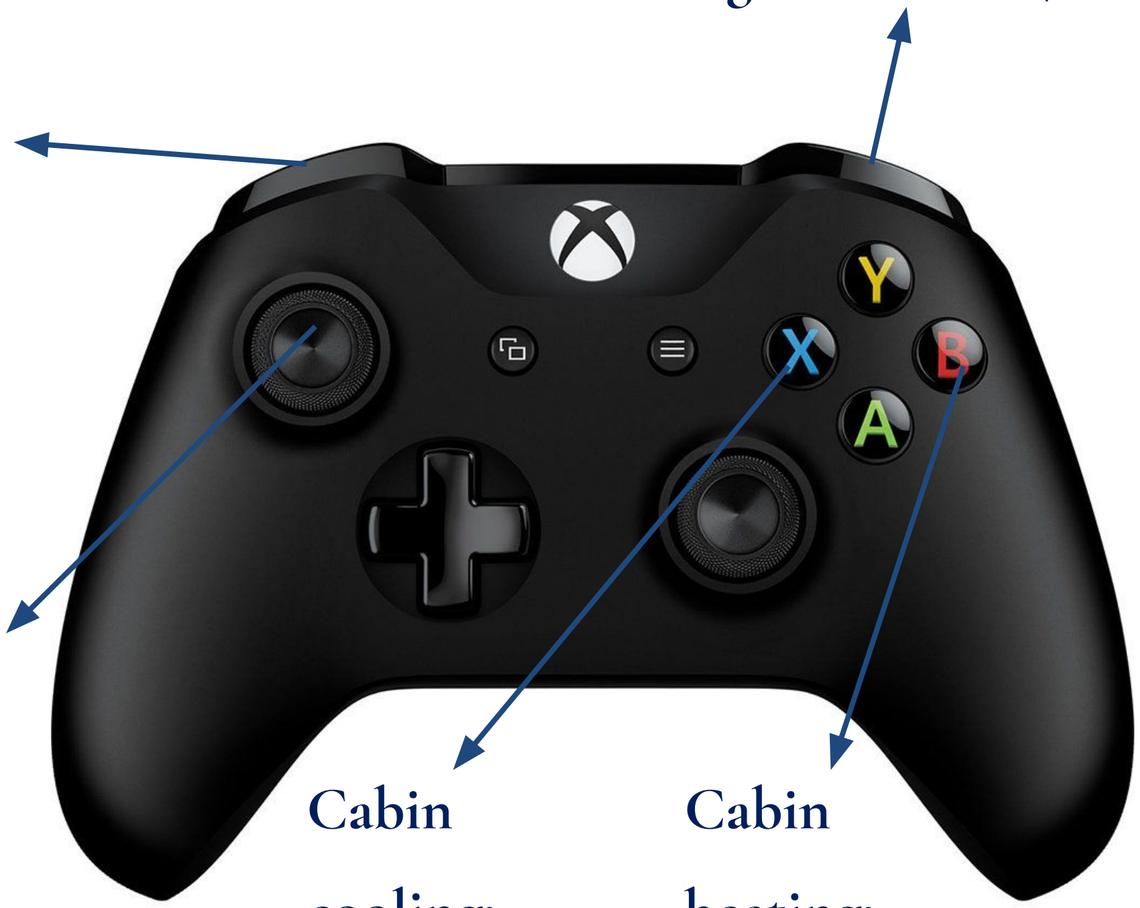
Controller Teleoperation

Headlights: Turn on/off

Brakes: Sends Brake Command to TSS

Linear & rotational motion: Moves Forward, Backward, Right, Left; Sends **Throttle** & **Steering** Command to TSS



Cabin cooling: Turn on/off

Cabin heating: Turn on/off

- **Move the rover** in the DUST simulation using XBOX controller
- Maps inputs to **intuitive controls**
- Control strength is **directly proportional** to how long it is held

HUMAN-IN-THE-LOOP (HITL) TESTING

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Overview of Testing Methodology

- **Comprehensive survey** administered to users via the Rice SONA program
- Users will test the user interface and will provide **feedback on the systems** via the
  - **NASA-TLX**
    - Measures the **perceived workload** of human-machine interface systems
    - Focuses on mental, physical and temporal demand; performance, effort and frustration
  - **UMUX-LITE**
    - Two-question metric that assesses a **system's usability & usefulness**
      - This system's capabilities meet my requirements.
      - This system is easy to use.

Subject Pool & Human Characteristics

- **Students** through SONA testing
  - Students receive extra credit through their participation
- Age Group: **18 - 50**
- **Easy access to many people**, with no limit on how many people we can test
- Aiming for **30+ participants**

RISKS, CHALLENGES AND MITIGATIONS

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Risk: TSS Connection Drops

TSS connection drops during test week

- **Risk:** TSS connection lost during EVA.
- **Impact:** Map, C&W, & Control lose live telemetry, impacting crew safety & operability.
- **Mitigation:**
  - Immediately notify user of connection loss, halt rover control.
  - Map and C&W persist from backend database.
  - Path planning uses last known PR position.
  - Resource estimation logic runs on cached data.
- **Our team will simulate TSS dropout scenarios during integration testing.**

Risk: AIA Hallucinations

AIA hallucinations on mission-critical data.

- **Risk**: AIA generates an incorrect response.
- **Impact**: Crew may act on misinformation, impacting EVA safety and task completion.
- **Mitigation**:
  - All telemetry values are injected directly into AIA context at query time.
  - If AIA still fails → backend falls back to hardcoded procedure templates w/TSS values
- **The team will conduct prompt testing to validate guardrails before test week.**

Risk: Autonomous Navigation & Resource Estimation Incomplete

Autonomous Navigation & Resource Estimation Incomplete by Test Week

- **Risk**: Autonomous path planning & resource estimation not done by test week.
- **Impact**: Partially demonstrated features, impacting competition scores.
- **Mitigation**:
  - Prioritize MVP with **straight-line best-path** planning and a **basic obstacle cost map**
  - Advanced optimizations based on DUST are stretch goals
  - Manual waypoint input as fallback for navigation during EVA

RESULTS, DEMO & FUTURE WORK

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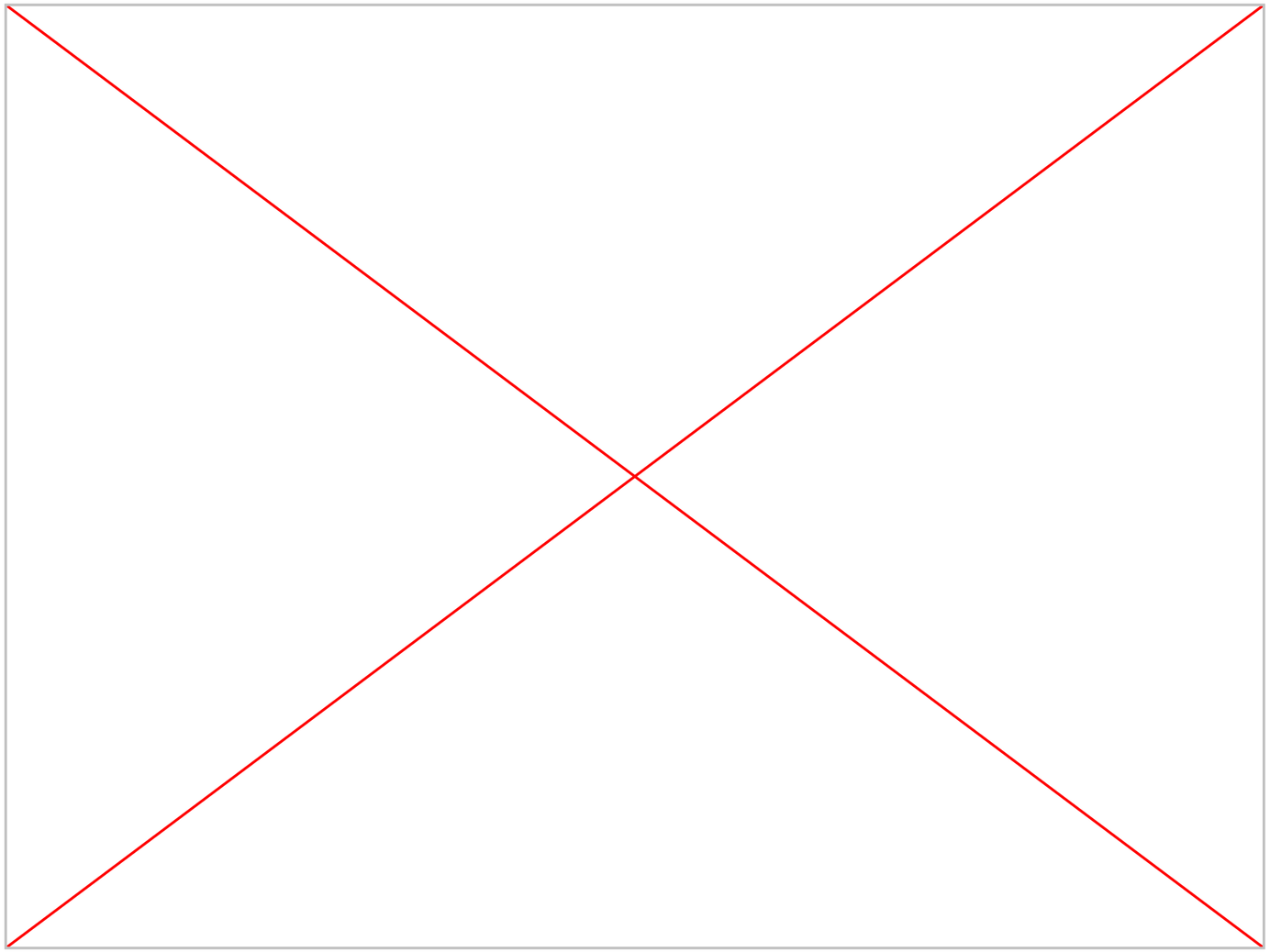
Demo



Future Work

- **Increase scope of project** to apply more closely to NASA mission
  - Our designs are **optimized for the specific telemetry datapoints** we receive through TSS
  - We understand that a real data dashboard screen would have 100+ datapoints that we would need to consider
- Implement **interoperability functionality**
  - The pressurized rover sole purpose is to assist the mission critical EVA tasks
- Quest 3 teleoperation using a VR digital twin of the DUST environment or custom-made VR client

THANK YOU

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